

2	
(a)	Work done per unit mass by an external agent in bringing a small point test mass from infinity to a point near Earth is negative.
	Gravitational potential at infinity is zero. It is negative because gravitational force is always attractive.
(b)	Negative gradient of gravitational potential - distance graph gives gravitational field strength.
	Tangent drawn with 2 points read off
	$\frac{(-0.075 - (-0.190)) \times 10^8}{(50 - 0) \times 10^6} = 0.23 \text{ N kg}^{-1}$ Gravitational force = $0.23 \text{ N kg}^{-1} \times 1200 \text{ kg} = 276 \text{ N}$
(c)	Gravitational force provides centripetal force
	Centripetal force = $m \frac{v^2}{r}$ $\frac{1}{2} m v^2 = \frac{\text{force} \times r}{2}$
	= $5.7 \times 10^9 \text{ J}$ (2sf)
(d)	Increase in GPE required = $\Delta \Phi \times m$ (= $6.36 \times 10^{10} \text{ J}$)
	Minimum energy required = Increase in GPE + KE at orbit
	= $6.93 \times 10^{10} \text{ J}$ (3 sf)
(e)	• Energy used to accelerate and decelerate the satellite to change direction of satellite • Energy to overcome resistive forces in Earth's atmosphere • Inefficiency in propulsion system • Energy required to launch satellite (due to rocket) • KE of satellite is not zero at Earth's surface due to Earth's rotation
3	